Intent Scoring Logic

How each visit and each visitor gets an intent score and a Low / Medium / High bucket. Written in plain language. Values are the V1 settings in `intent_engine/config.py`.

1. The big picture

Every visit to the store (a **session**) is turned into a score from 0 to 100. The score is then turned into a bucket. A visitor's own intent is the score and bucket of their **best** session.

Step	What happens
1	Skip sessions that should not be scored (bots, empty visits).
2	Add up points for what the visitor did, each activity capped.
3	Subtract penalties for bounces and frustration.
4	Multiply by how close to buying the visitor got (page-type multiplier), then limit to 0-100.
5	Turn the score into Low / Medium / High using the brand's own traffic.
6	Force the operational bucket to High if the visitor showed hard buying actions.
7	Visitor's intent = their best-scoring session.

2. Sessions that are not scored

These sessions get no score and are marked *excluded*. They are never counted as Low.

- **Bots and crawlers**: for example Googlebot, Bingbot, curl, wget, Selenium, Puppeteer, Playwright, headless Chrome, and any browser name containing the word "bot".
- **Empty visits**: a single event lasting one second or less.

3. Points for behaviour

Each activity earns points up to a maximum count. Doing more than the cap earns nothing extra, so no single behaviour can dominate.

Activity	Points each	Counts up to	Most it can add
Product pages viewed	11	4	44
Useful clicks	2.2	12	26.4
Minutes on site	4.5	6 minutes	27
Scrolls	1.8	8	14.4
Pages viewed	1.5	6	9

Product views are worth the most because looking at products is the strongest buying signal. The maximum possible is 120.8, which is why the final score is capped at 100.

4. Penalties

Penalty	Points	When it applies
Bounce	-20	The visitor did 2 or fewer things, or left in under 5 seconds.
Struggle	-12	The visitor clicked 10 or more times and the share of clicks that did nothing ("dead clicks") is above that brand's 90th percentile. This suggests frustration. If the brand's 90th percentile is not known yet, the penalty is skipped.

5. Page-type multiplier

After penalties, the score is multiplied by a factor based on the most commercial page the visitor reached.

Furthest page reached	Multiplier
Checkout	x 1.15
Product page	x 1.05
Collection page	x 0.95
Home page	x 0.85
Anything else	x 0.70

Final session score = (behaviour points + penalties) x multiplier, then limited to the range 0 to 100.

Worked example. A visitor views 3 product pages (33), makes 6 useful clicks (13.2), spends 4 minutes (18), scrolls 5 times (9) and views 5 pages (7.5). Behaviour points: 80.7. No penalties. Reached a product page: x 1.05. **Score: about 84.7.**

6. Buckets: Low, Medium, High

Buckets are **relative to each brand's own traffic**, not fixed score lines. Once a day (02:00 IST) the engine looks at that brand's scored sessions from the previous day and sets two cut-offs.

Bucket	Which sessions
Low	Bottom 35% of scores (below the 35th percentile)
Medium	Next 37% (35th to 72nd percentile)
High	Top 28% (72nd percentile and above)

Not enough data. If a day has fewer than 500 scored sessions, percentiles are not reliable. The engine then uses fixed cut-offs: under 25 is Low, 25 to under 55 is Medium, 55 or more is High.

The cut-offs used for each day are saved (table `intent_thresholds_daily`), so a day's buckets can be explained later.

7. Hard override (operational bucket)

Some behaviour is a fact, not a guess. If any one of these is true, the session's **operational bucket** is set to High whatever its score:

- The visitor **landed on a checkout page**.
- The visitor **started checkout**.
- The visitor **added something to the cart**.

The override changes only the operational bucket. The predictive score and predictive bucket are never altered. So each session carries two labels: the **predictive bucket** (from the score) and the **operational bucket** (with the override).

8. Visitor-level intent

A visitor (actor) can have many sessions. Their intent is the **highest-scoring scored session**, with that session's bucket. The number of scored sessions is stored too. A visitor whose sessions were all excluded has no score at all; they are never defaulted to Low. The visitor record is recalculated in full whenever one of their sessions is newly scored or re-bucketed, never kept as a running maximum.

9. When it runs

Job	Schedule	What it does
Scoring	Every 2 hours	Scores newly closed sessions (with a 1-hour overlap) and updates the visitors they belong to.
Calibration	Daily, 02:00 IST	Computes the day's Low/Medium/High cut-offs from yesterday's sessions and re-buckets that day.
Rollups	Every 15 minutes	Builds the daily dashboard summaries from the ingested events.

10. Things to know

- The page-type multipliers, the fallback cut-offs (25 and 55) and the bot list are **unvalidated V1 placeholders**: sensible guesses, not yet calibrated against real conversion data.
- Because buckets are percentile-based, roughly 28% of a brand's scored sessions will always be High, whatever the absolute scores. A brand with weak traffic still shows a High group. Use the score and the operational bucket together for absolute judgments.
- Calibration works on "yesterday" only. After a data backfill, older days keep the cut-offs they had, or the fallback cut-offs, until a one-off recalibration is run.
- All weights, caps, penalties, multipliers and thresholds live in one file, `intent_engine/config.py`.